

# National University of Computing and Emerging Sciences - CFD Campus



Due Date: May 10, 2026

**Software for mobile and devices      Assignment#04**

**Marks: 100**

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## **Submission Guidelines:**

1. Do not submit your assignment after the deadline. Late submission will not be accepted.
2. **Plagiarism from internet (ChatGPT) or any peer is strictly prohibited.**
3. **In case of plagiarism zero marks will be awarded.**
4. Submit your compressed project file in the format: RollNo1\_RollNo2\_RollNo3\_A4 (23F-1234\_23F-5678\_A4).
5. Deadline: May 10, 2026

### Overview

In Assignment #03, you integrated remote data services and local relational databases into your Android application. For Assignment #04, you will elevate your application by adopting industry-standard cloud technologies, modern UI paradigms, and proactive user engagement mechanisms. This assignment focuses on four core pillars: Firebase Cloud Services, Push Notifications, Jetpack Compose UI, and GitHub Collaboration. Additionally, you are required to innovate by implementing at least two new functionalities beyond what has been covered in class, demonstrating your ability to independently research and integrate advanced Android capabilities.

### Objective

Enhance your existing Android application by integrating Firebase backend services, implementing a notification system, modernizing portions of the UI using Jetpack Compose, maintaining collaborative version control via GitHub, and introducing independently researched features that demonstrate exceptional Android development skills.

### Global Constraints (Strict Requirements)

- **Existing Codebase:** All tasks must be implemented within the project you developed for Assignment #03.
- **Threading:** All Firebase operations and background tasks must be executed asynchronously using Kotlin Coroutines or Firebase callbacks to prevent UI freezing.
- **Firebase SDK:** You must use the official Firebase Android SDK. Direct REST calls to Firebase endpoints are not permitted.
- **Modular UI:** Continue using Fragments for existing content screens. Jetpack Compose must be used for at least one new screen or component as specified in the requirements.

### Functional Requirements

| ID | Component                              | Requirement Description                                                                                                                                                                                                                                                                  |
|----|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F1 | Firebase Authentication                | Integrate Firebase Authentication into your app. Implement at least two sign-in methods (e.g., Email/Password and Google Sign-In). Ensure user sessions persist across app restarts and provide appropriate UI for login, registration, and logout flows.                                |
| F2 | Firebase Firestore / Realtime Database | Migrate or extend your data layer to use Firebase Firestore or Realtime Database. Implement real-time data synchronization so that changes made on one device are immediately reflected on other connected devices. Maintain at least two collections/nodes with a logical relationship. |

## GitHub Collaboration Requirement

- **Shared Repository:** All groups must maintain their work on the same GitHub repository used in previous assignments.
- **Measurable Contribution:** Every group member must have clear, descriptive commits showing their implementation of Firebase, Notifications, Compose, or the self-researched features.
- **Branching Strategy:** Each major feature (Firebase, FCM, Compose, New Features) must be developed on a separate branch and merged into main via a Pull Request with a peer review comment.
- **Version Control:** Marks will be awarded based on the frequency and quality of collaborative commits, proper branch management, and documented merge practices.

## Logic Map Documentation

A **logic\_map.pdf** is mandatory. It must clearly link the requirements above to your code logic, and for self-researched features, must include references to official documentation or tutorials used.

Map Structure:

| Requirement ID | Class / File Name  | Function / Method  | Implementation Description                                                |
|----------------|--------------------|--------------------|---------------------------------------------------------------------------|
| F1             | AuthRepository.kt  | signInWithGoogle() | Handles Google OAuth token exchange with Firebase Auth.                   |
| F2             | FirestoreHelper.kt | syncUserData()     | Listens for real-time updates on the 'users' collection and refreshes UI. |

## Submission Requirements

1. Zipped Android Studio Project (Perform "Clean Project" before zipping).
2. Logic Map (PDF) — including research references for self-implemented features.
3. Link to GitHub Repository with all feature branches and Pull Requests visible.
4. PowerPoint Presentation (PPT) — a concise slide deck demonstrating your app's new features, architecture decisions, and a brief walkthrough of the self-researched functionalities.

## Grading Rubric

| Evaluation Criteria                                           | Marks      |
|---------------------------------------------------------------|------------|
| Firebase Integration (Authentication + Firestore/Realtime DB) | 20         |
| Clean code and Viva                                           | 15         |
| Logic Map Accuracy                                            | 10         |
| GitHub Collaboration                                          | 10         |
| New Functionality #1 & #2 (Self-Researched Features)          | 25         |
| Exceptional Handling, Edge Cases & Complete Testing           | 10         |
| PowerPoint Presentation (PPT)                                 | 10         |
| <b>Total Marks</b>                                            | <b>100</b> |

**Good Luck 😊**

